

Pratibha

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EDUCATION

Rochester Institute of Technology

Master of Science, Data Science

GPA: 3.8 / 4.0

Aug 2024 – May 2026

EXPERIENCE

IpserLab

Software Engineer

California

Aug 2025 – Dec 2025, June 2026 – Present

- Analyzed real-time system performance metrics including latency, throughput, and error rates, delivering structured findings to engineering and product stakeholders to support infrastructure decision-making.
- Improved data retrieval accuracy by identifying and correcting field mapping inconsistencies in **Apache Lucene**, reducing search errors and aligning backend data structures with evolving reporting requirements.

IonSoft

Data Scientist

North Carolina

Aug 2022 – Aug 2024

- Engineered behavioral features from transaction and user interaction data using **SQL** and **Python**, building **XGBoost** classification models with **SHAP** explainability that drove a **28%** improvement in anomaly detection rate over existing rule-based systems for enterprise clients.
- Conducted statistical analysis and exploratory data analysis across large multi-source datasets to identify trends, assess risk exposure, and deliver structured recommendations that informed strategic decisions for client stakeholders across fintech and healthcare verticals.
- Built end-to-end **ELT** workflows on **Databricks** using **PySpark** and **Delta Lake**, reducing pipeline failures by **30%** through automated data quality checks and ensuring full data lineage traceability across client environments.
- Designed and deployed **Tableau** and **Power BI** dashboards tracking operational KPIs, model performance, and business metrics, enabling cross-functional leadership teams to make faster, data-informed decisions and reducing manual reporting effort.
- Led end-to-end project delivery across **Agile** sprint cycles, defining scope, goals, and deliverables with cross-functional client teams and translating complex analytical outputs into clear business insights for both technical and non-technical audiences.

Rochester Institute of Technology

Research Assistant — Narrative Dynamics & Attention Shift Analysis

Rochester, New York

Jan 2026 – May 2026

- Built an **NLP** pipeline on **GCP** processing **400,000+** YouTube news videos across **20+** global channels using NER and keyword extraction, producing temporal narrative snapshots — contributing to an ongoing research publication on media narrative dynamics.
- Revealed U.S. channels averaged **54%** political video ratio versus **4%** for international outlets using Jaccard similarity and volatility metrics, identifying instability spikes during the 2016, 2020, and 2024 election cycles.

SKILLS

Languages: Python, SQL, Java, R, MATLAB

Data Engineering: Apache Spark, PySpark, Airflow, Databricks, Delta Lake, Snowflake, dbt, PostgreSQL, Kafka

Cloud & Platforms: AWS, Azure, GCP, Docker, Kubernetes, Git, CI/CD

AI & ML: XGBoost, Random Forest, NLP, LLMs, LangChain, LangGraph, Prompt Engineering, RAG Pipelines, TensorFlow, PyTorch, Scikit-Learn

Analytics & BI: Exploratory Data Analysis, Statistical Analysis, A/B Testing, KPI Development, Forecasting, Root Cause Analysis

Visualization: Tableau, Power BI, Streamlit, Microsoft Excel, Matplotlib, Seaborn, Jupyter

Certifications: Google Data Analytics Professional Certificate, AWS Certified

PROJECTS

Real-Time Financial Fraud Detection System | PySpark, Hadoop, Python, Spark MLlib, LangChain

- Built a large-scale analytics pipeline on **1M+** transactions comparing batch vs. near real-time detection workflows, training **XGBoost** and **Random Forest** models achieving **92%** accuracy with precision, recall, and F1-score optimization.
- Integrated **LLM**-based transaction narrative summarization via **LangChain** with output quality checks to auto-flag suspicious patterns, delivering analyst-ready explanations for flagged transactions at scale.

SupportOps Ticket Intelligence | Python, SQL, Scikit-Learn, Streamlit

- Analyzed **8K+** support tickets using **SQL** and **Python** to compute KPIs including SLA breach rate, resolution latency, and backlog growth, with regression-based root cause analysis explaining **45%** of turnaround-time variance.
- Built a logistic regression classifier achieving **0.78 ROC-AUC** to predict SLA breaches, delivering results via a **Streamlit** dashboard for interactive KPI monitoring by non-technical stakeholders.

Scare-O-Meter | Python, Random Forest, LangChain, BioRadio, Signal Processing | 1st Place, AWARE-AI Hackathon 2026

- Designed a **Random Forest** biosignal classification pipeline achieving **87%** accuracy for gesture recognition under live competition constraints using EOG and EMG signals through **BioRadio** hardware.
- Integrated **LLM**-based stress estimation via **LangChain** to generate real-time physiological summaries, enabling automated adaptive content delivery based on user emotional state.